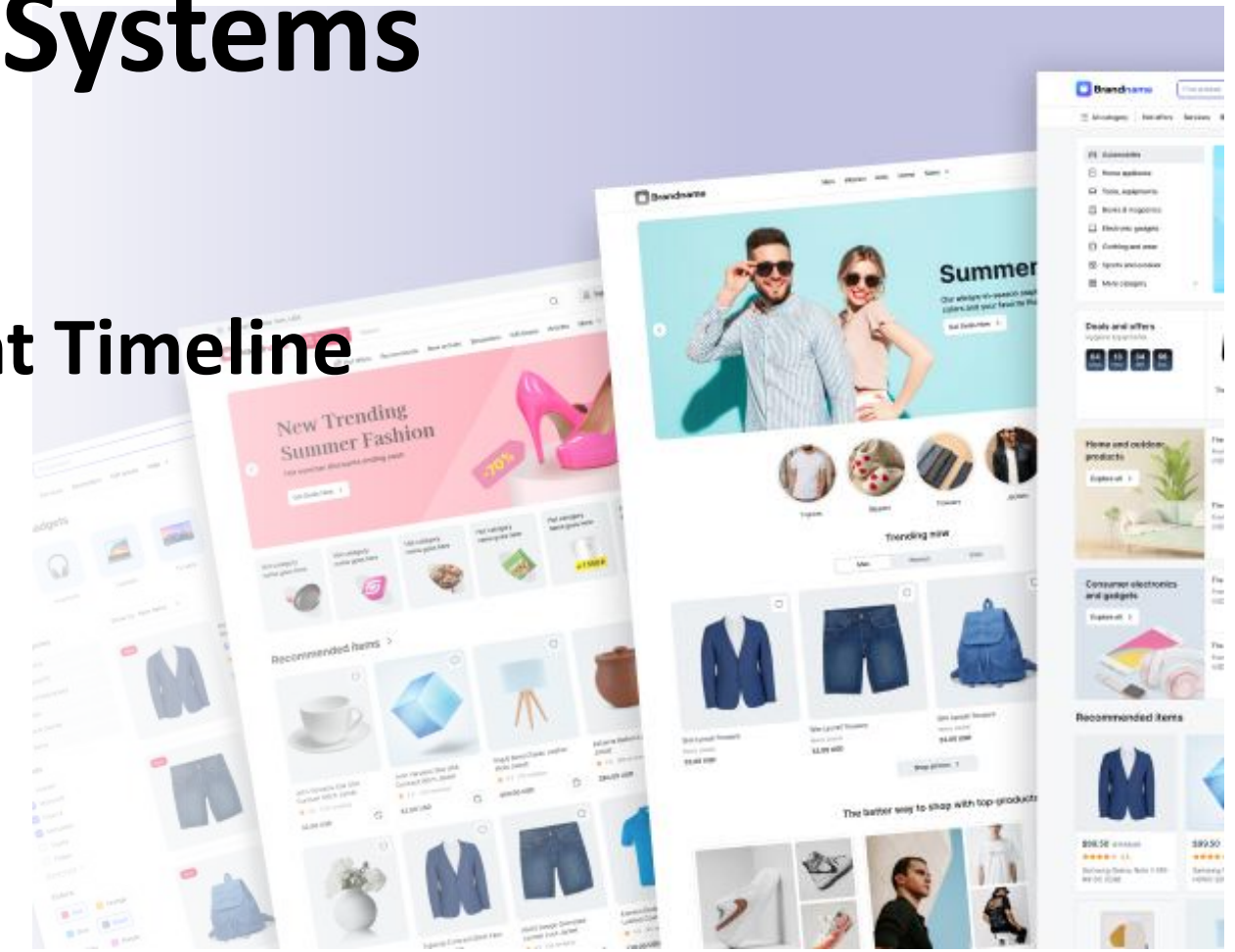
ICT2142 E-Business Systems

Week 01
E-Commerce Site Development Timeline



Phase 1 Planning & Requirements
(Weeks 1–2)

Expected Learning Outcomes

After completing this project, students should be able to:

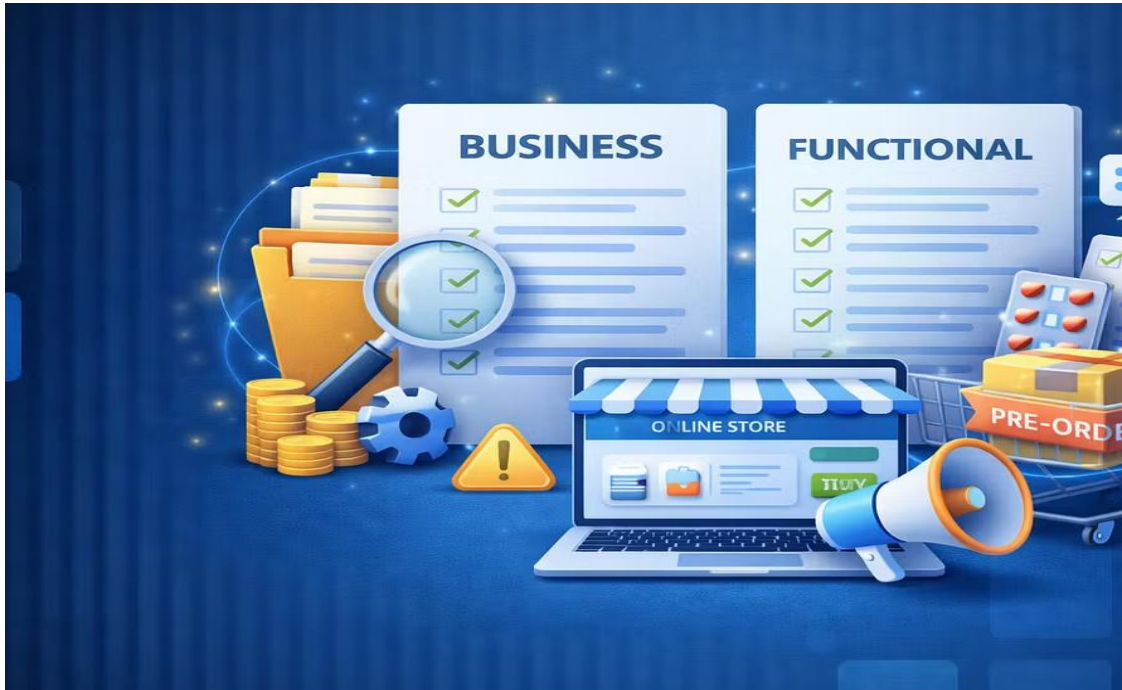
- Design a responsive E-Commerce website interface
- Implement essential E-Commerce functionalities
- Develop a structured relational database
- Simulate or integrate an online payment workflow
- Apply security mechanisms for web applications
- Demonstrate understanding of real-world online business workflows



Objective of the Mini Project

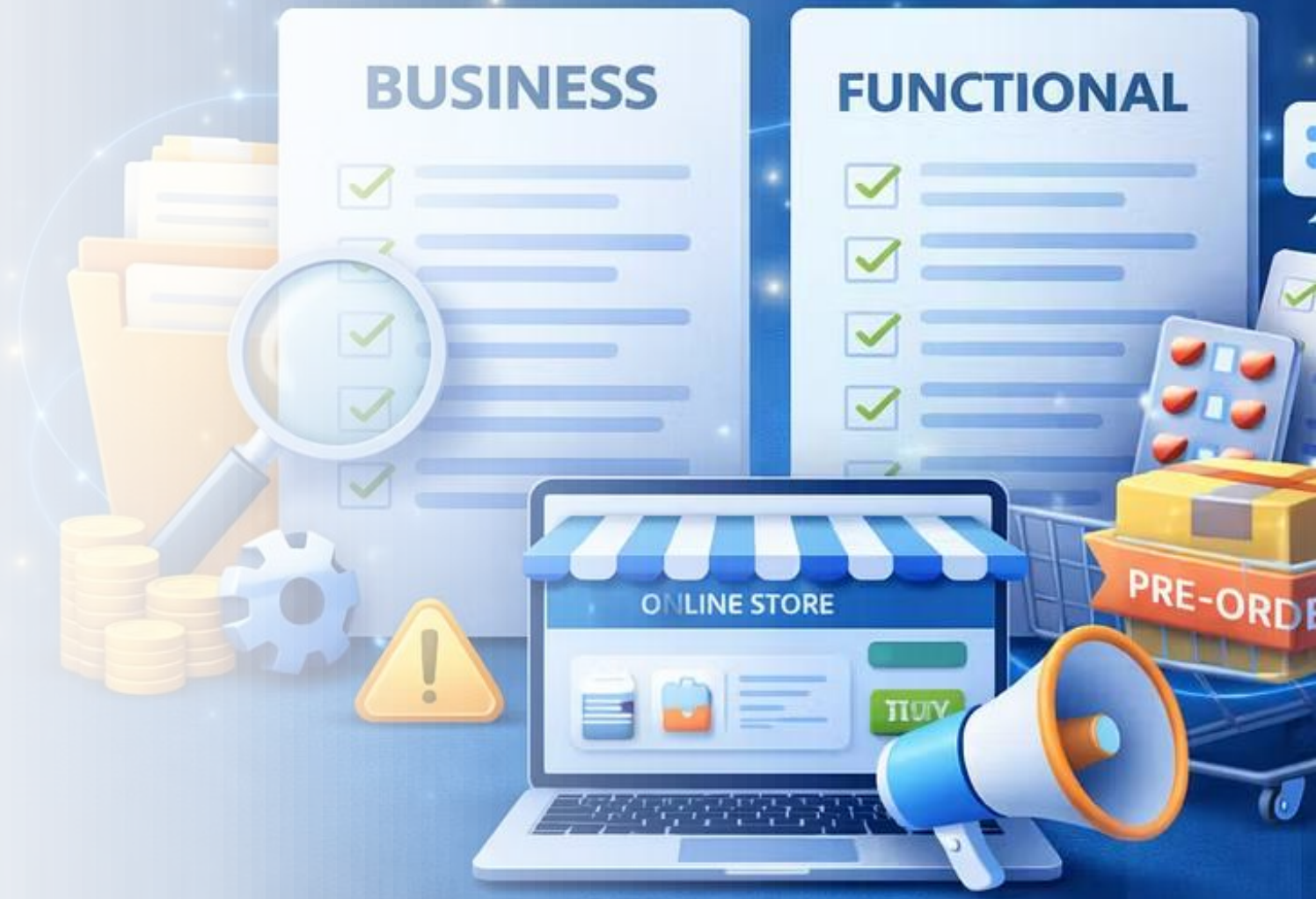
The objective of this mini project is to evaluate students' **practical knowledge and technical skills** in designing and developing a functional **E-Commerce website** for a selected online business. Students are expected to apply concepts related to:

- Web interface design
- Database development
- Online payment workflow
- Security implementation
- Core E-Commerce functionalities



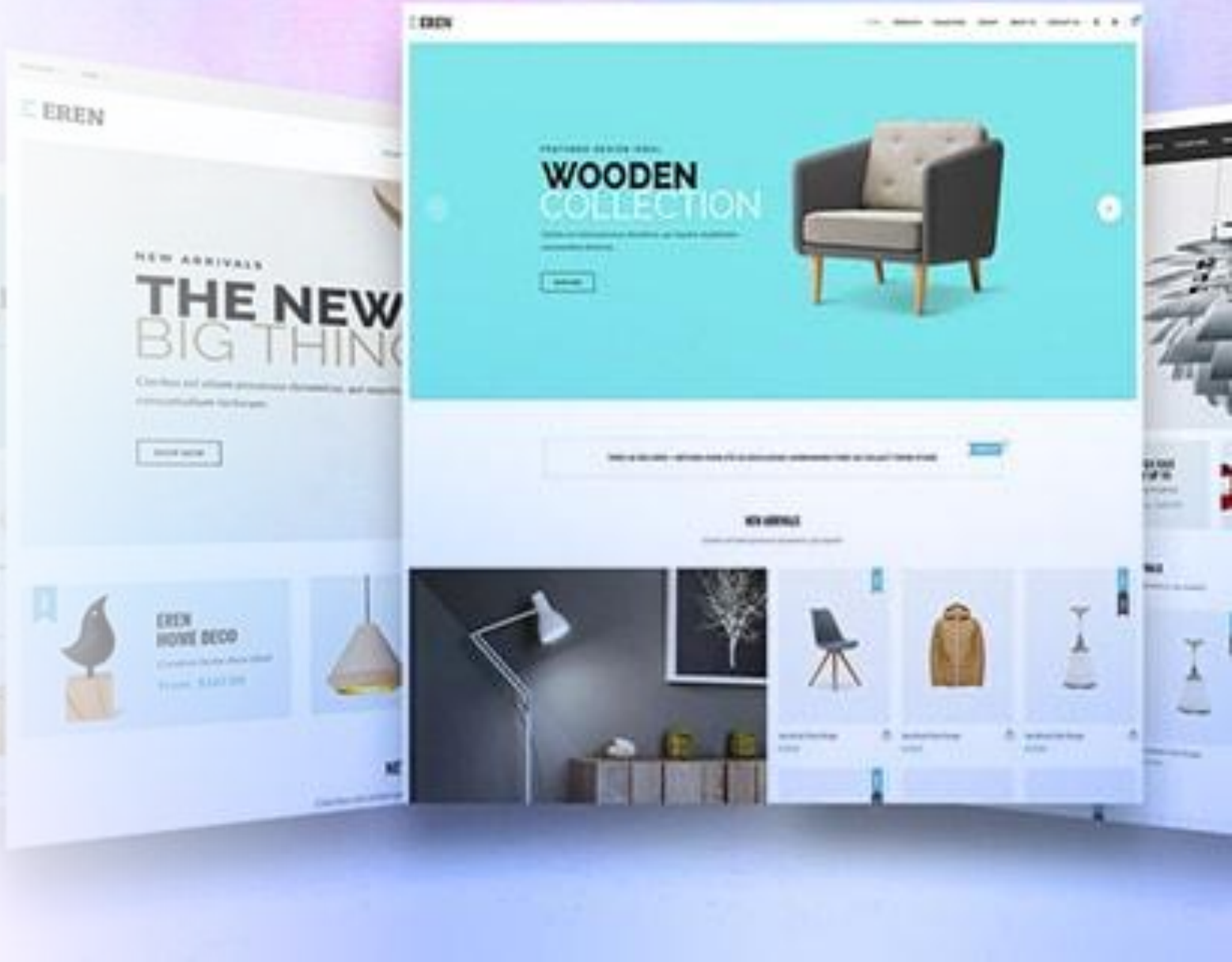
Week 1: Project Initiation & Requirement Gathering

Goal: Define "**What**" you are building and "**Who**" you are building it for.



1. Define the Product Function (The Store Concept)

- Should pick a specific type of online store
  - Examples:
    - Boutique clothing store
    - Handmade crafts shop
    - Organic grocery store
    - Digital bookstore.
- **Why it matters:** A focused function helps determine what features are necessary
  - Examples:
    - A clothing store needs size and color variations
    - A digital bookstore needs downloadable file links



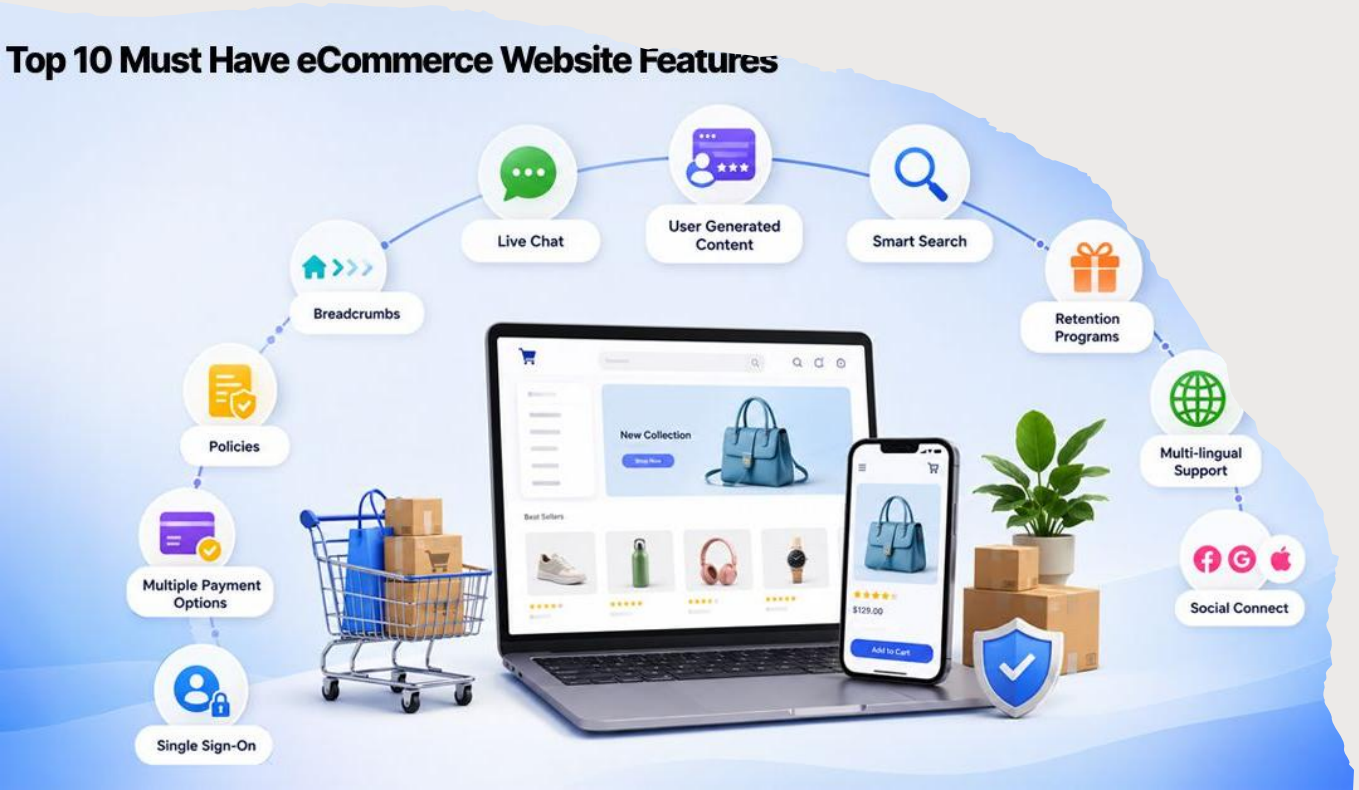
2. Identify Target Users

- Tasks: Briefly outline who will use the website.
  - Example:
    - Young people buying clothes, or Busy parents buying groceries

- **Why it matters:** User behavior dictates the design choices, font sizes, and checkout simplicity.



3. List Core Features (Functional Requirements)



- Students must separate "must-have" features from "nice-to-have" features
  - Must-Haves:
    - Product catalog (viewing items)
    - Search and filter (by category or price)
    - Shopping cart (add, remove, update quantity)
    - User authentication (sign up, log in, log out)
    - Checkout process (shipping address and order placement)
  - Nice-to-Haves (Only if time permits later):
    - Product reviews and ratings
    - Wishlists
    - Live chat support

4. Choose the Technology Stack

- Based on their current coding skill level, students should select their tools:
    - Custom Development (For learning fundamental web design.)
        - HTML/CSS/JavaScript
    - Modern Full-Stack
        - React.js/Next.js for frontend
        - Node.js or Python for backend
        - MongoDB or PostgreSQL for the database.



Report Submission



At the end of each practical session student shout submit completed section as a part of the report submission.

Project Report (Recommended Structure)

The report should include:

- Introduction
- Business description
- Requirement analysis
- System architecture
- Database design
- UI screenshots
- Security features implemented
- Payment workflow explanation
- Testing results
- Conclusion

Report Submission Requirement for Week 01

At the end of the week 01 practical session

Students should submit below sections as a word document as a part of the report with the given link in the LMS.

The report should include:

- Introduction
- Business description
- Requirement analysis
- System architecture

